

Sovol SV02 firmware Installation headache

I figured I would post this for anyone else that ever finds themselves in this same predicament.

1: go to and find the firmware for your printer. Mine was the dual extruder SV02. Download the firmware package.

[Sovol Firmware page](<https://sovol3d.com/pages/download>)

2: Extract the folder as a whole file with everything in it somewhere like the desktop.

3: !!!!!IMPORTANT!!!!!

RENAME THE FOLDER SO THAT IT DOES NOT HAVE ANY CHINESE CHARACTERS OR ANY OTHER OPERATION WILL FAIL!!!!

I just renamed it by delete it the characters.

4: Download AVRDUDESS. Set it up and get it running.

[AVRDUDESS](<https://blog.zakkemle.net/avrdudess-a-gui-for-avrdude/>)

consider donating, it's not mine but it's helped a lot over the years and it's going to help you

5: after you have AVRDUDESS open and running:

1:On the top right where it says MCU select: ATmega2560

2:Right under that where it says Presets select:
Arduino Mega (ATmega2560)

3:Top left where it says Programmer:
It should say: ATmel STK500 Version (insert version number)

4: BAUD rate has to be 115200

5: Make sure the printer is switched off. Plug in the usb cable into the printer and then to the computer. !!! IMPORTANT!! MAKE SURE NO OTHER PROGRAM THAT USES A USB COM PORT IS OPEN!!!! I.E Cura, Repetier...

6: Find the COM port the printer is in. I found it using Device Manager.

7:Now that you know what COM port your printer is on. Select it in AVRDUDESS.

8:Under the Programmer section, you will see one called Flash. Select the .hex file for your printer in the folder that you downloaded from the SOVOL website. This file is found in the folder that you renamed and it has a .hex file extension. Mine for example was for a mixing extruder

without BLtouch. Remember how I said to rename it and remove any chines characters? If you don't do this, everything will fail and give you an error saying it couldn't read the file.

9: Leave every other setting alone unless it was stated to change it. The prep work is done.

6: Now that you have everything set up, on the bottom left towards the middle of the screen you will see a Program button. Click it. ITS IMPORTANT THAT YOU DONT MESS WITH THE PRINTER OR CABLE OR ANYTHING UNTIL AVRDUDE SAYS ITS DONE!!!

7: If AVRDUDE says it's done and there were no errors. Congrats!!!! You just flashed fresh firmware into your printer and can unplug it and turn it on normally. Plug the power back in and flip the switch. The screen should load the progress bar and you can print happily ever after.

7.2: If AVRDUDE gives you a SYNC error or not responding error: I learned this trick a while ago when messing with Arduinos and all that fun stuff. Unplug the printer. Move your mouse over the program button but don't press it. What you're going to do is plug in the printer and about one(1) second later you're going to click the Program button. This resets the atmega and let's the computer read and write to it. This may take a little trial and error and a tiny bit more or less time between the plug in and button press but it works. If all works and is good, AVRDUDE will tell you that the device is initialized and ready to accept instructions. Congratulations, now refer to 7:.

If all works out, you now have a useable printer again and can go on with printing. This also works for other boards and printers. The board that Sovol uses happens to be a Creality V2.2.1 or V2.2 which is also in Ender printers so it can work for those as well.